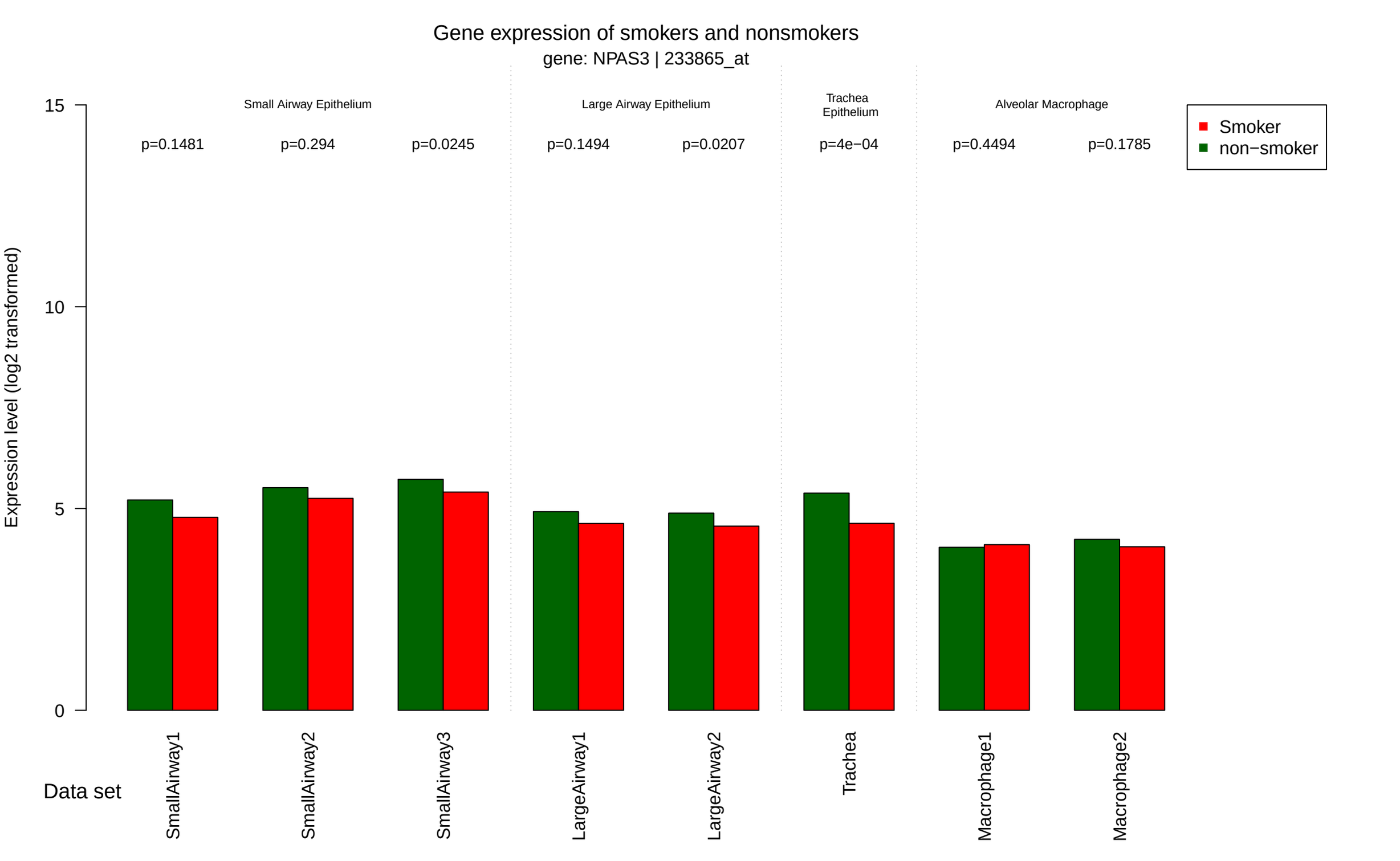
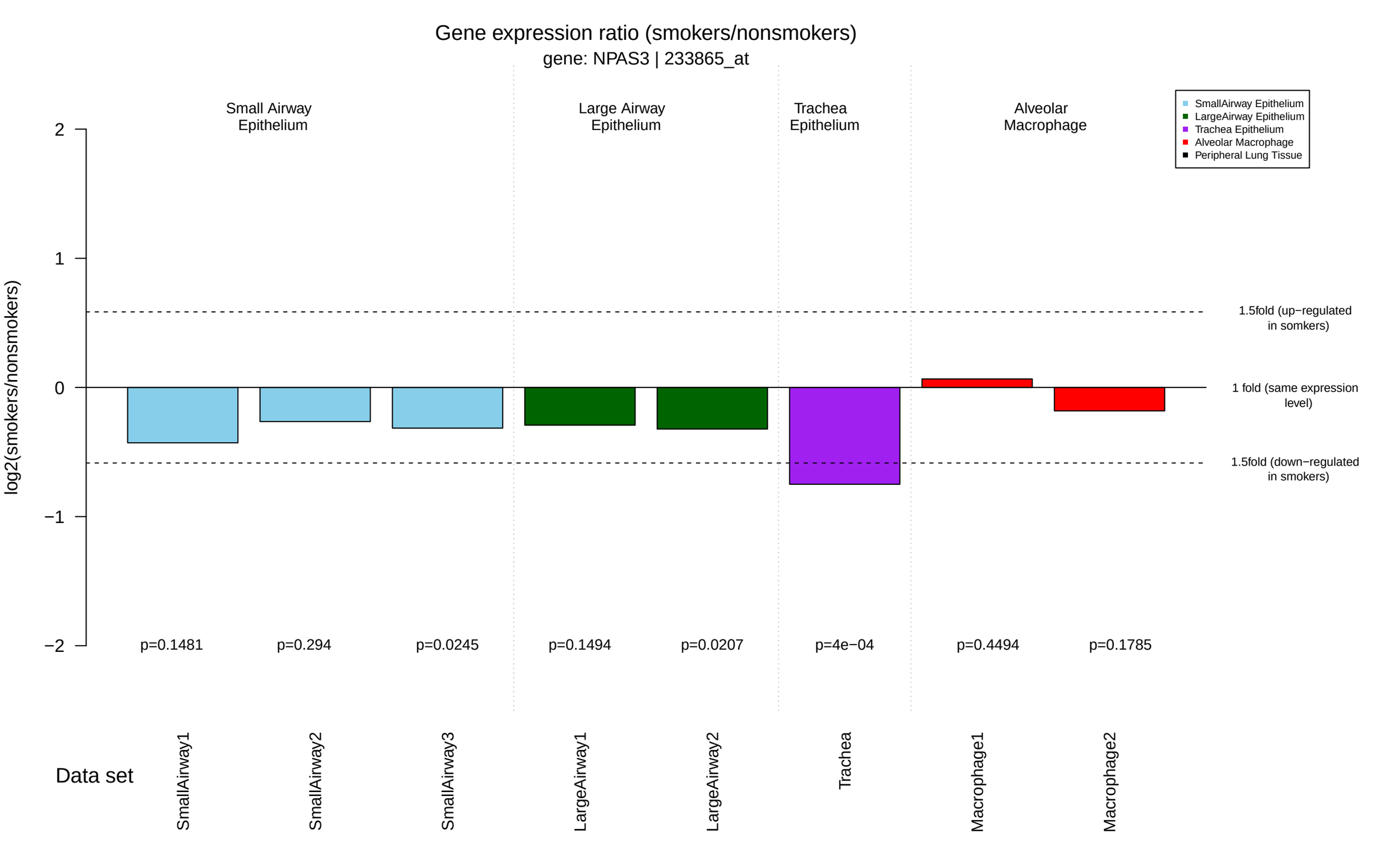
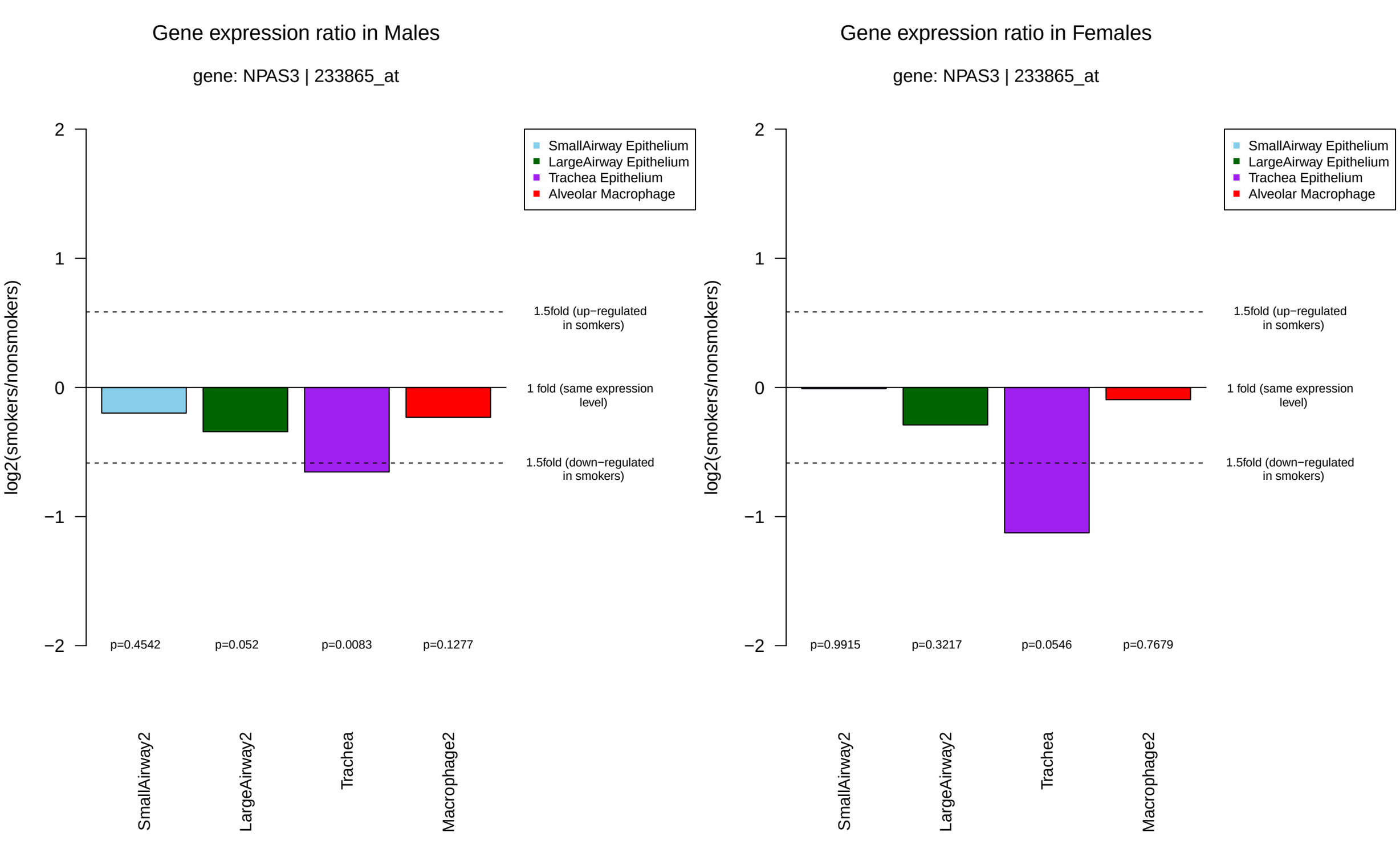
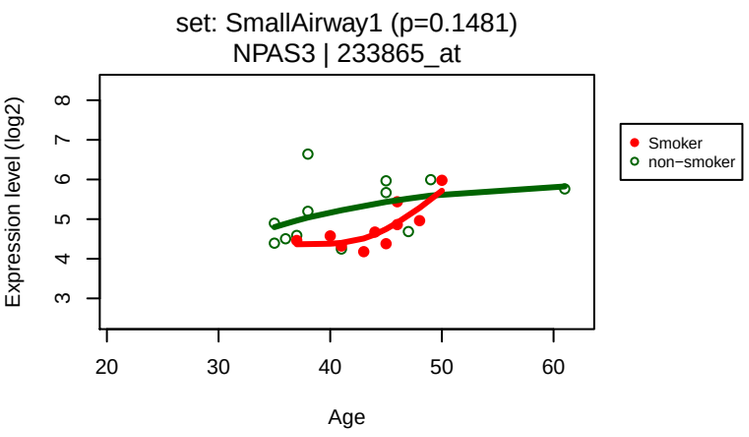
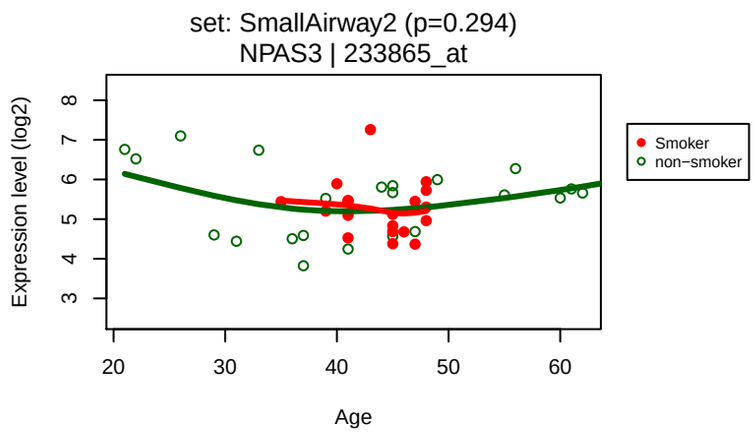
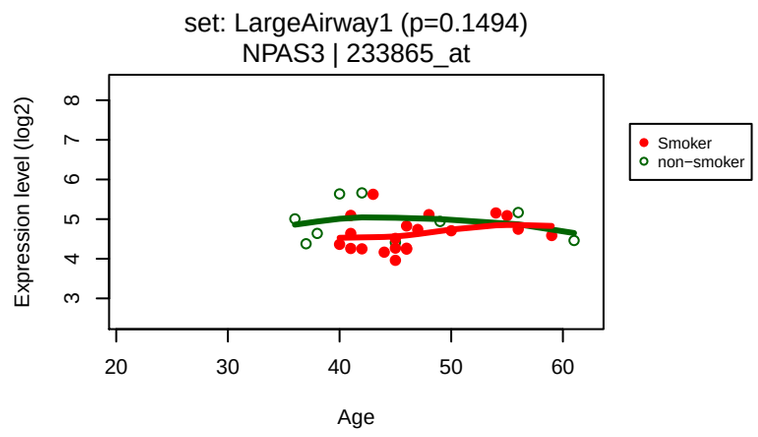

Gene expression of smokers and nonsmokers
gene: NPAS3 | 233865_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1481
p=0.294
p=0.0245
p=0.1494
p=0.0207
p=4e−04
p=0.4494
p=0.1785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

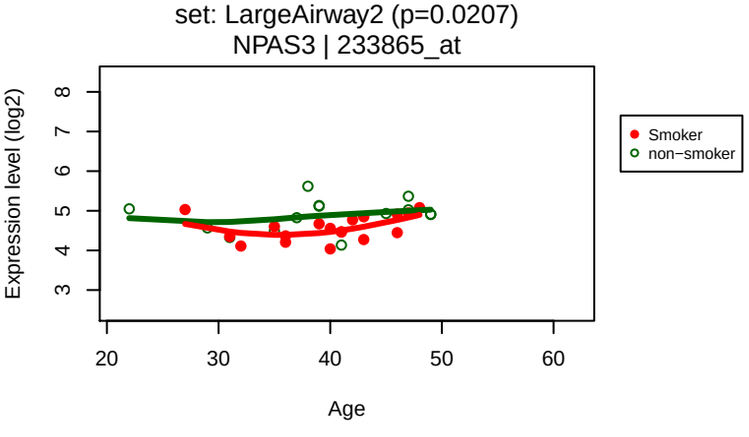

Gene expression ratio (smokers/nonsmokers)
gene: NPAS3 | 233865_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1481
p=0.294
p=0.0245
p=0.1494
p=0.0207
p=4e−04
p=0.4494
p=0.1785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

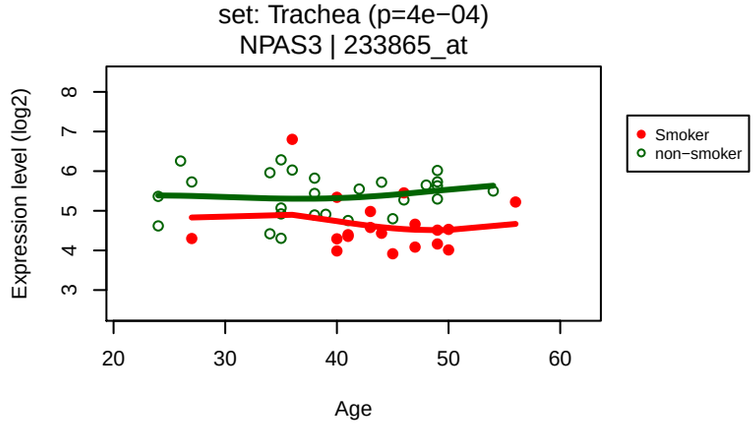

Gene expression ratio in Males
gene: NPAS3 | 233865_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4542
p=0.052
p=0.0083
p=0.1277
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS3 | 233865_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9915
p=0.3217
p=0.0546
p=0.7679
SmallAirway2
LargeAirway2
Trachea
Macrophage2

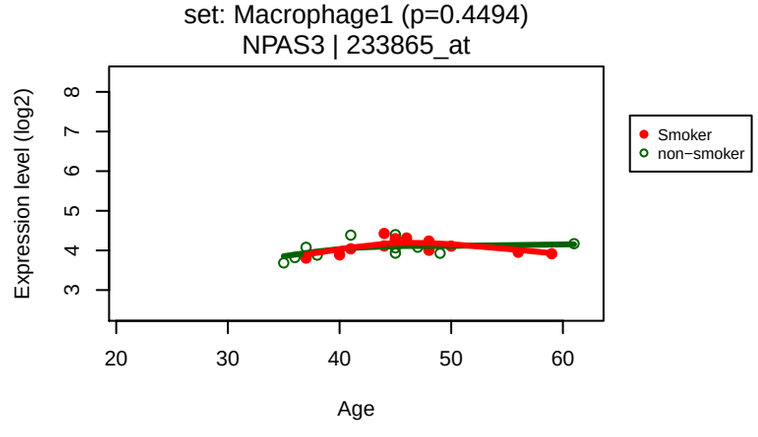

set: SmallAirway1 (p=0.1481)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker

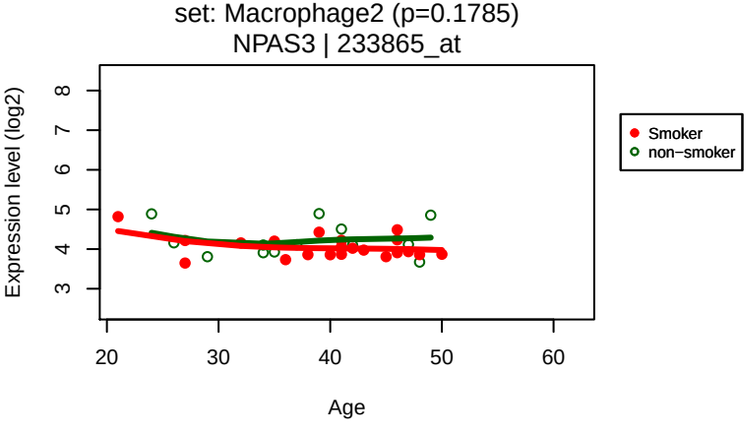

set: SmallAirway2 (p=0.294)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker

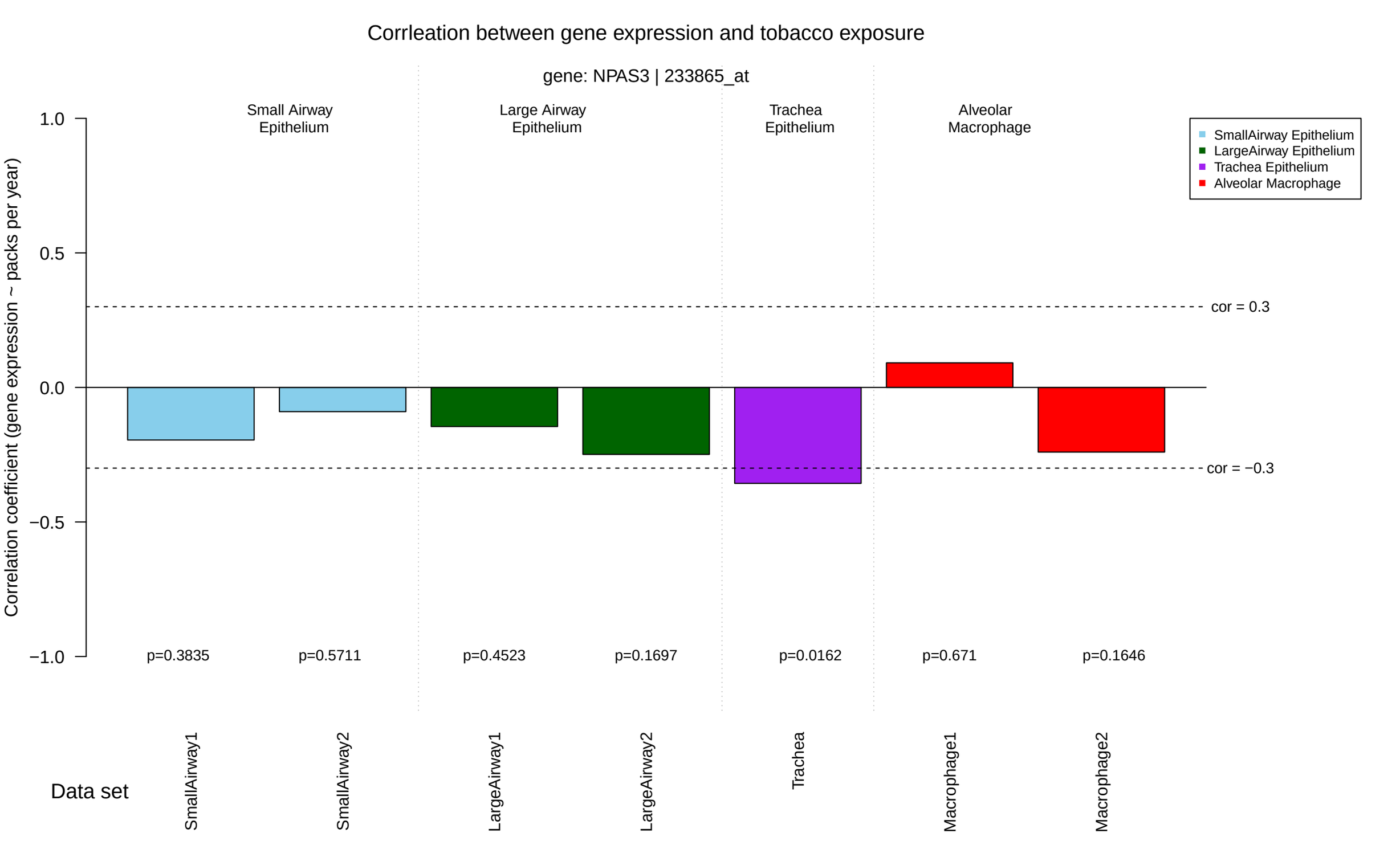

set: LargeAirway1 (p=0.1494)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0207)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4494)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1785)
NPAS3 | 233865_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS3 | 233865_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3835
p=0.5711
p=0.4523
p=0.1697
p=0.0162
p=0.671
p=0.1646
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2